Disciplina:

DEVOPS

Professor:
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Terraform

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Objetivos

Ao final deste módulo o aluno deverá estar apto a:

Utilizar o Terraform para gerenciamento de infraestrutura.

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Justificativa

O conhecimento do funcionamento da ferramenta propicia a utilização adequada.

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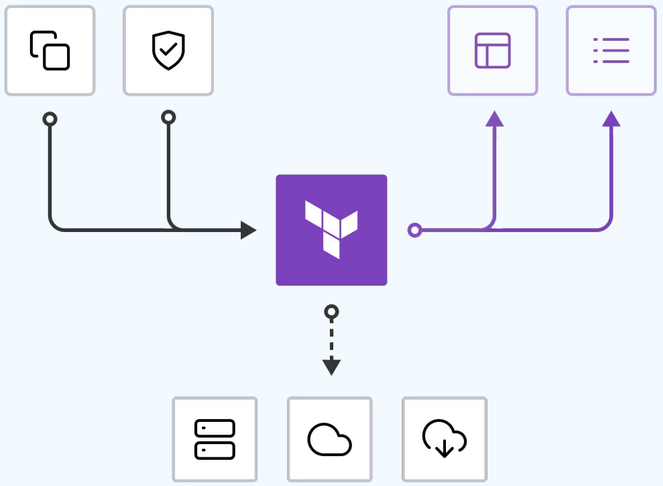
O que é Terraform

O HashiCorp Terraform é uma ferramenta de infraestrutura como código que permite definir recursos em nuvem e locais por meio de arquivos de configuração legíveis para humanos, que podem ser versionados, reutilizados e compartilhados.

Com um fluxo de trabalho consistente, a solução permite provisionar e gerenciar toda a infraestrutura ao longo de seu ciclo de vida.



https://developer.hashicorp.com/terraform

Por que Terraform?

Gerenciamento de qualquer infraestrutura por meio de providers (Terraform Registry).

Configurações padronizadas por meio de módulos.

Colaboração por meio de uso de VCS e do HCP Terraform.

Controle da infraestrutura, por meio de gerenciamento de estados (state files).

Automação de mudanças.

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Providers

Providers are a logical abstraction of an upstream API. They are responsible for understanding API interactions and exposing resources.

AWS

Azure

Google Cloud Platform

Kubernetes

Alibaba Cloud

Oracle Cloud Infrastructure

Active Directory
by: hashicorp

Archive
by: hashicorp

Assert
by: hashicorp

AWS Cloud Control
by: hashicorp

Azure Active Directory
by: hashicorp

Azure Stack
by: hashicorp

https://registry.terraform.io/browse/providers

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Modules

Modules are self-contained packages of Terraform configurations that are managed as a group.

terraform-aws-modules / iam

Terraform module to create AWS IAM resources 🇺🇦

2 days ago 205.2M aws provider

terraform-aws-modules / vpc

Terraform module to create AWS VPC resources 🇺🇦

25 days ago 113.1M aws provider

terraform-aws-modules / s3-bucket

Terraform module to create AWS S3 resources 🇺🇦

25 days ago 102.6M aws provider

terraform-aws-modules / eks

Terraform module to create Amazon Elastic Kubernetes (EKS) resources 🇺🇦

2 days ago 85.9M aws provider

https://registry.terraform.io/browse/providers

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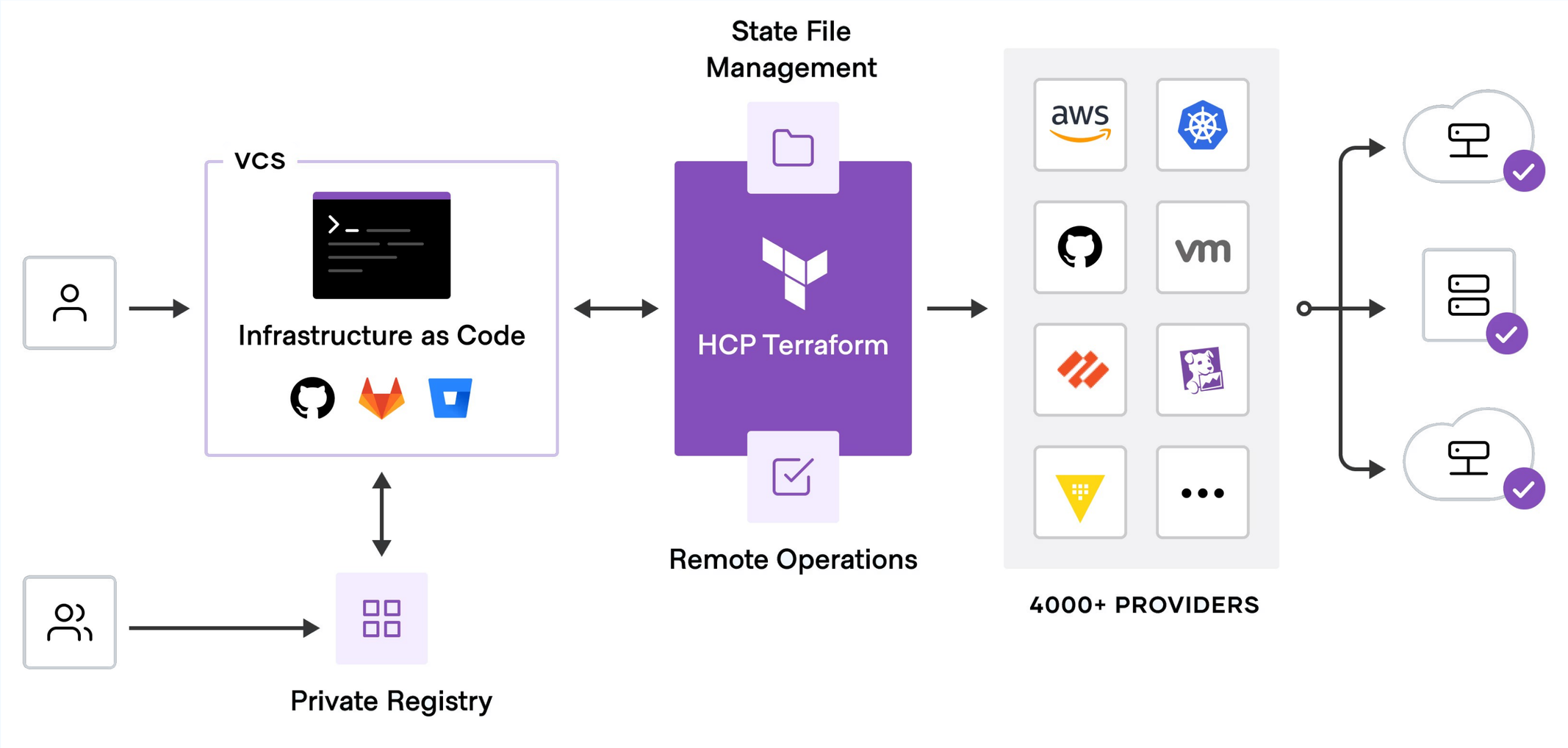

https://developer.hashicorp.com/terraform/tutorials/cloud-get-started/cloud-sign-up

Quando Usar Terraform

Gerenciamento de múltiplos ambientes.

Gerenciamento de infraestrutura complexa.

Gerenciamento de infraestrutura multicloud.

Automação da implantação de infraestrutura.

Quando Não Usar Terraform

Infraestrutura de servidor único

Alternativas: Ansible e Puppet.

Infraestrutura como serviço (IaaS)

Alternativas: AWS CloudFormation ou Azure Resource Manager.

Plataforma como serviço (PaaS)

Alternativas: Netlify e Google App Engine

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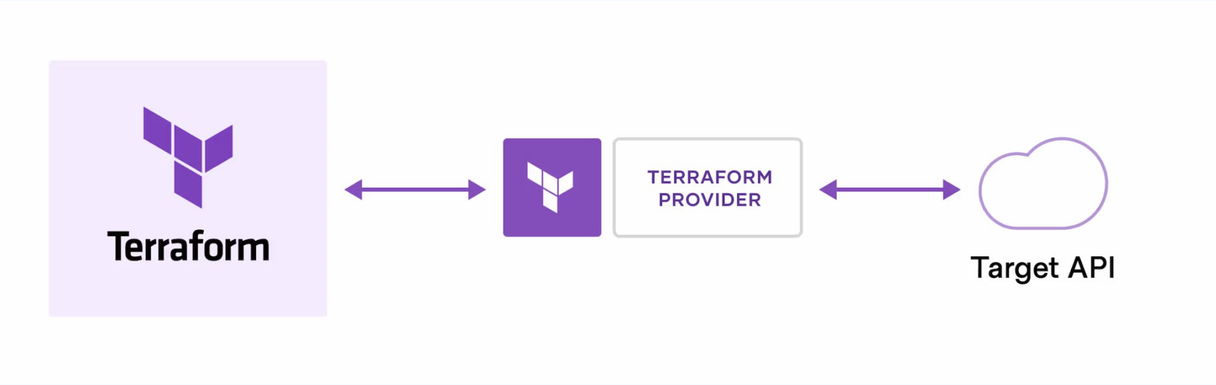
Como o Terraform Funciona

O Terraform cria e gerencia recursos em plataformas de nuvem e outros serviços por meio de suas interfaces de programação de aplicativos (APIs).

Os provedores permitem que o Terraform funcione com praticamente qualquer plataforma ou serviço que possua uma API acessível.



https://developer.hashicorp.com/terraform

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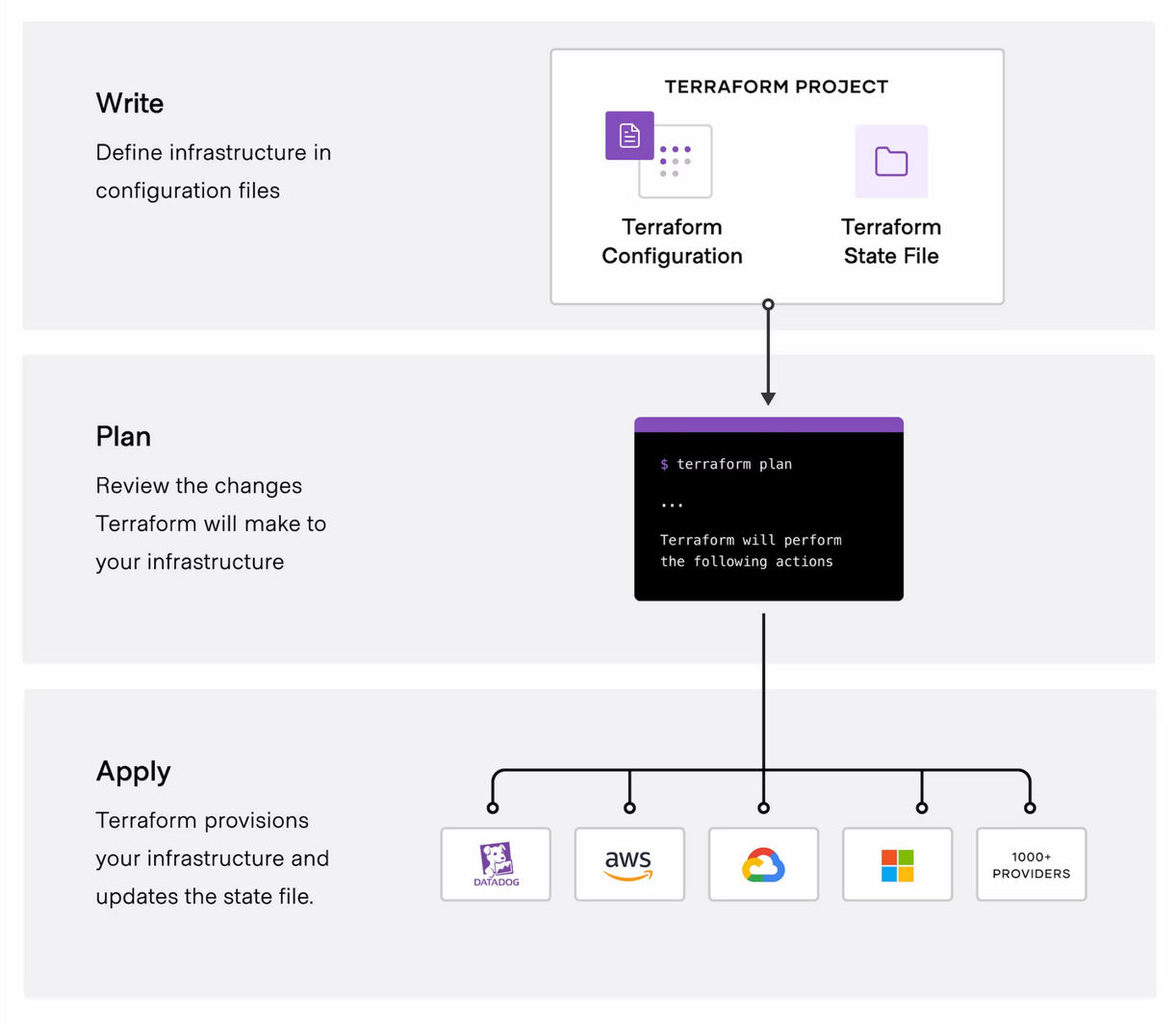
Fluxo de Trabalho



https://developer.hashicorp.com/terraform

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Definição de providers

Permite definir os providers e respectivas versões.

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 4.16"
    }
  }

  required_version = ">= 1.2.0"
}

provider "aws" {
  region  = "us-east-1"
}
```

terraform.tf

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Inicializar

terraform init

Quando uma nova configuração é criada é necessário inicializar o diretório.

A inicialização faz o download e instalação dos providers definidos.

```
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 4.16"...
- Installing hashicorp/aws v4.67.0...
- Installed hashicorp/aws v4.67.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

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Escrever

O conjunto de arquivos usados para descrever a infraestrutura no Terraform é conhecido como configuração.

Cada configuração do Terraform deve residir em um diretório próprio.

```
resource "aws_instance" "cyo_web_server" {
  ami           = "ami-05b10e08d247fb927"
  instance_type = "t2.micro"
  subnet_id     = aws_subnet.cyo_subnet.id
  security_groups = [aws_security_group.cyo_sg.id]

  user_data = <<-EOF
              #!/bin/bash
              yum update -y
              yum install -y httpd
              systemctl start httpd
              systemctl enable httpd
              echo "<h1>Hello Conchayoro</h1>" > /var/www/html/index.html
              EOF

  tags = {
    Name = "cyo_web_server"
  }
}
```

main.tf

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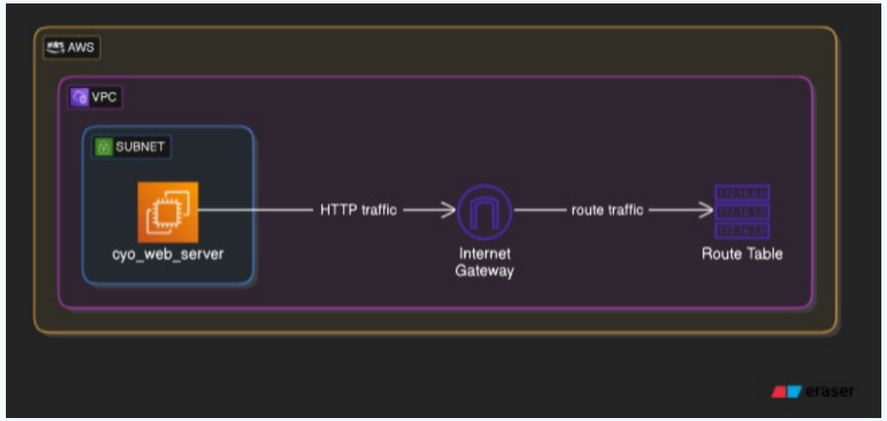
```
resource "aws_vpc" "cyo_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "cyo_subnet" {
  vpc_id            = aws_vpc.cyo_vpc.id
  cidr_block        = "10.0.1.0/24"
  availability_zone = "us-east-1a"
}

resource "aws_internet_gateway" "cyo_ig" {
  vpc_id = aws_vpc.cyo_vpc.id
}

resource "aws_route_table" "cyo_rt" {
  vpc_id = aws_vpc.cyo_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.cyo_ig.id
  }
}

resource "aws_route_table_association" "cyo_rta" {
  subnet_id      = aws_subnet.cyo_subnet.id
  route_table_id = aws_route_table.cyo_rt.id
}
```

```
resource "aws_security_group" "cyo_sg" {
  vpc_id = aws_vpc.cyo_vpc.id

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

resource "aws_instance" "cyo_web_server" {
  ami           = "ami-05b10e08d247fb927"
  instance_type = "t2.micro"
  subnet_id     = aws_subnet.cyo_subnet.id
  security_groups = [aws_security_group.cyo_sg.id]
  associate_public_ip_address = true

  user_data = <<-EOF
              #!/bin/bash
              yum update -y
              yum install -y httpd
              systemctl start httpd
              systemctl enable httpd
              echo "<h1>Hello Conchayoro</h1>" > /var/www/html/index.html
              EOF

  tags = {
    Name = "cyo_web_server"
  }
}
```



main.tf

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Planejar

terraform plan

Terraform cria um plano de execução descrevendo a infraestrutura que será criada, atualizada ou destruída com base na infraestrutura existente e na configuração.

```
Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.cyo_web_server will be created
  + resource "aws_instance" "cyo_web_server" {
      + ami                                  = "ami-05b10e08d247fb927"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
```

```
Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_id = (known after apply)
  + public_ip   = (known after apply)
  + website_url = (known after apply)

──────────────────────────────────────────────────────────────────────────────
```

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Aplicar

terraform apply

Terraform executa as operações propostas na ordem correta, respeitando as dependências entre os recursos.

Por exemplo, se forem alteradas as propriedades de uma VPC e for alterado o número de máquinas virtuais nessa VPC, o Terraform irá atualizar a VPC antes.

```
Terraform used the selected providers to generate the following execution plan
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.cyo_web_server will be created
  + resource "aws_instance" "cyo_web_server" {
      + ami                                  = "ami-05b10e08d247fb927"
```

```
aws_vpc.cyo_vpc: Creating...
aws_vpc.cyo_vpc: Creation complete after 6s [id=vpc-09de89fd3a3b66f3b]
aws_internet_gateway.cyo_ig: Creating...
aws_subnet.cyo_subnet: Creating...
aws_security_group.cyo_sg: Creating...
aws_subnet.cyo_subnet: Creation complete after 3s [id=subnet-0b8ef30b96a58a00f]
aws_internet_gateway.cyo_ig: Creation complete after 3s [id=igw-0b4a58f0ddf383da
d]
aws_route_table.cyo_rt: Creating...
aws_route_table.cyo_rt: Creation complete after 3s [id=rtb-0f799b40fe98a38d5]
aws_route_table_association.cyo_rta: Creating...
aws_security_group.cyo_sg: Creation complete after 6s [id=sg-0a3327b5afd27e0ff]
aws_instance.cyo_web_server: Creating...
aws_route_table_association.cyo_rta: Creation complete after 1s [id=rtbassoc-016
cea5c2291ff039]
aws_instance.cyo_web_server: Still creating... [10s elapsed]
aws_instance.cyo_web_server: Creation complete after 14s [id=i-09eb22d9bd56fa32f
```

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Inspecionar

terraform show

```
# aws_instance.cyo_web_server:
resource "aws_instance" "cyo_web_server" {
    ami                                  = "ami-05b10e08d247fb927"
    arn                                  = "arn:aws:ec2:us-east-1:947392381371:instance/i-01e95ea7f8f035f8d"
    associate_public_ip_address          = true
    availability_zone                    = "us-east-1a"
    cpu_core_count                       = 1
    cpu_threads_per_core                 = 1
    disable_api_stop                     = false
    disable_api_termination              = false
    ebs_optimized                        = false
    get_password_data                    = false
    hibernation                          = false
    host_id                              = null
    iam_instance_profile                 = null
    id                                   = "i-01e95ea7f8f035f8d"
    instance_initiated_shutdown_behavior = "stop"
    instance_state                       = "running"
    instance_type                        = "t2.micro"
    ipv6_address_count                   = 0
    ipv6_addresses                       = []
    key_name                             = null
    monitoring                           = false
    outpost_arn                          = null
    password_data                        = null
    placement_group                      = null
    placement_partition_number           = 0
    primary_network_interface_id         = "eni-0a74d346ab123014e"
    private_dns                          = "ip-10-0-1-172.ec2.internal"
    private_ip                           = "10.0.1.172"
    public_dns                           = null
    public_ip                            = "3.87.67.64"
```

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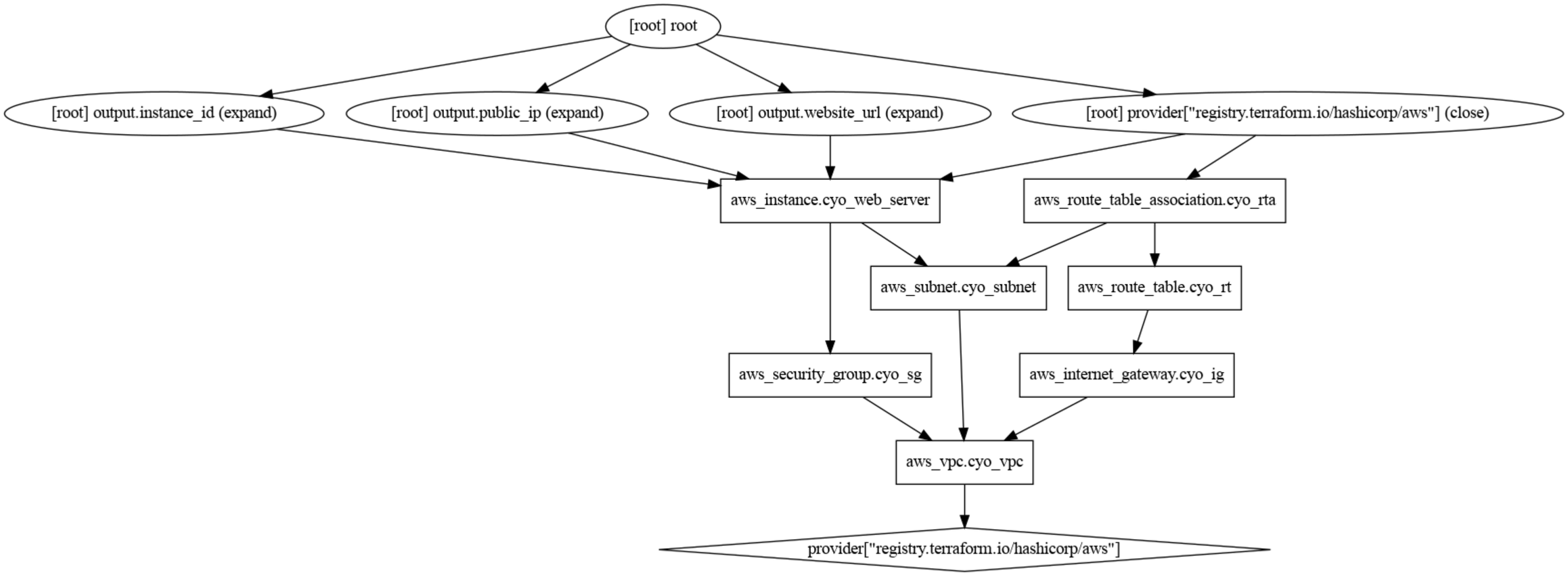
Inspecionar

terraform graph -type=plan | dot -Tpng >graph.png



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Destruir

terraform destroy

```
aws_vpc.cyo_vpc: Refreshing state... [id=vpc-09de89fd3a3b66f3b]
aws_internet_gateway.cyo_ig: Refreshing state... [id=igw-0b4a58f0ddf383dad]
aws_subnet.cyo_subnet: Refreshing state... [id=subnet-0b8ef30b96a58a00f]
aws_security_group.cyo_sg: Refreshing state... [id=sg-0a3327b5afd27e0ff]
aws_route_table.cyo_rt: Refreshing state... [id=rtb-0f799b40fe98a38d5]
aws_instance.cyo_web_server: Refreshing state... [id=i-01e95ea7f8f035f8d]
aws_route_table_association.cyo_rta: Refreshing state... [id=rtbassoc-016cea5c22
91ff039]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.cyo_web_server will be destroyed
  - resource "aws_instance" "cyo_web_server" {
      - ami                                  = "ami-05b10e08d247fb927" -> null
```

```
aws_route_table_association.cyo_rta: Destroying... [id=rtbassoc-016cea5c2291ff039]
aws_instance.cyo_web_server: Destroying... [id=i-01e95ea7f8f035f8d]
aws_route_table_association.cyo_rta: Destruction complete after 4s
aws_route_table.cyo_rt: Destroying... [id=rtb-0f799b40fe98a38d5]
aws_route_table.cyo_rt: Destruction complete after 1s
aws_internet_gateway.cyo_ig: Destroying... [id=igw-0b4a58f0ddf383dad]
aws_instance.cyo_web_server: Still destroying... [id=i-01e95ea7f8f035f8d, 10s elapsed]
aws_internet_gateway.cyo_ig: Still destroying... [id=igw-0b4a58f0ddf383dad, 10s elapsed]
aws_instance.cyo_web_server: Still destroying... [id=i-01e95ea7f8f035f8d, 20s elapsed]
aws_internet_gateway.cyo_ig: Still destroying... [id=igw-0b4a58f0ddf383dad, 20s elapsed]
aws_instance.cyo_web_server: Still destroying... [id=i-01e95ea7f8f035f8d, 30s elapsed]
aws_internet_gateway.cyo_ig: Still destroying... [id=igw-0b4a58f0ddf383dad, 30s elapsed]
aws_instance.cyo_web_server: Still destroying... [id=i-01e95ea7f8f035f8d, 40s elapsed]
aws_internet_gateway.cyo_ig: Still destroying... [id=igw-0b4a58f0ddf383dad, 40s elapsed]
aws_instance.cyo_web_server: Still destroying... [id=i-01e95ea7f8f035f8d, 50s elapsed]
aws_internet_gateway.cyo_ig: Still destroying... [id=igw-0b4a58f0ddf383dad, 50s elapsed]
aws_instance.cyo_web_server: Destruction complete after 56s
aws_subnet.cyo_subnet: Destroying... [id=subnet-0b8ef30b96a58a00f]
aws_security_group.cyo_sg: Destroying... [id=sg-0a3327b5afd27e0ff]
aws_internet_gateway.cyo_ig: Destruction complete after 52s
aws_subnet.cyo_subnet: Destruction complete after 2s
aws_security_group.cyo_sg: Destruction complete after 3s
aws_vpc.cyo_vpc: Destroying... [id=vpc-09de89fd3a3b66f3b]
aws_vpc.cyo_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
```

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Pontos chave

O Terraform é uma ferramenta de IaC que permite gerenciar infraestruturas locais e em nuvem por meio de código versionável e reutilizável. Ele automatiza mudanças, controla estados e facilita a colaboração.

Quando Usar? Infraestruturas complexas, multicloud e com múltiplos ambientes.
Quando Não Usar? Servidor único (use Ansible/Puppet).
IaaS (use AWS CloudFormation/Azure Resource Manager).
PaaS (use Netlify/Google App Engine).

O Terraform interage com APIs para gerenciar recursos via provedores.
Fluxo de Trabalho: terraform init → Instala provedores. Escrever configurações.
terraform plan → Planeja mudanças. terraform apply → Aplica mudanças.

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Referências Bibliográficas

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Kerim Satirli, Taylor Dolezal. **Terraform Cookbook**. O'Reilly Media, Inc. 2024.

Yevgeniy Brikman. Terraform: Up and Running, 3rd Edition. O'Reilly Media, Inc. 2022.

Hashcorp. **Terraform**. Disponível em https://developer.hashicorp.com/terraform. Acessao em fevereiro/2025.

Fim de Apresentação!!!

Obrigado pela atenção!!!

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